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Project Number: #6

Project Name: OpenCL Linear Regression

Project Report #6

1. What machine you ran this on

Rabbit server

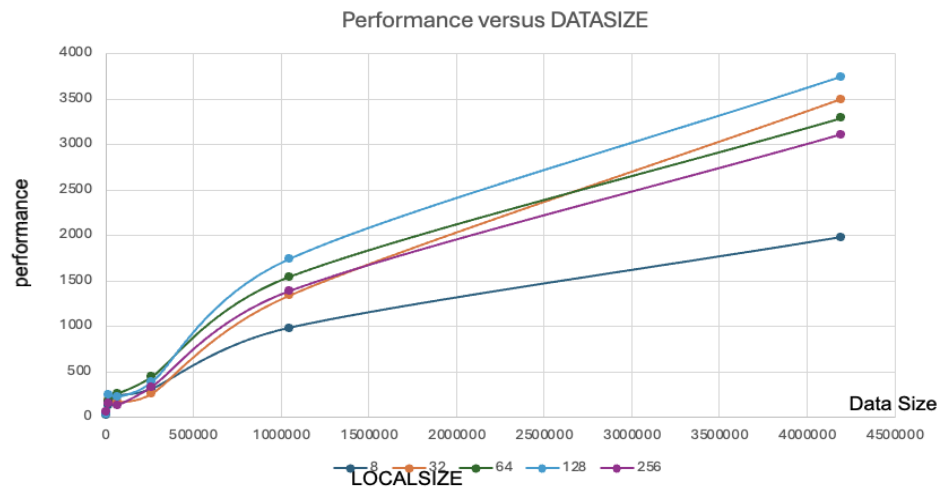
2. Determined the correct M and B values

M= 5.01 B = 7.00

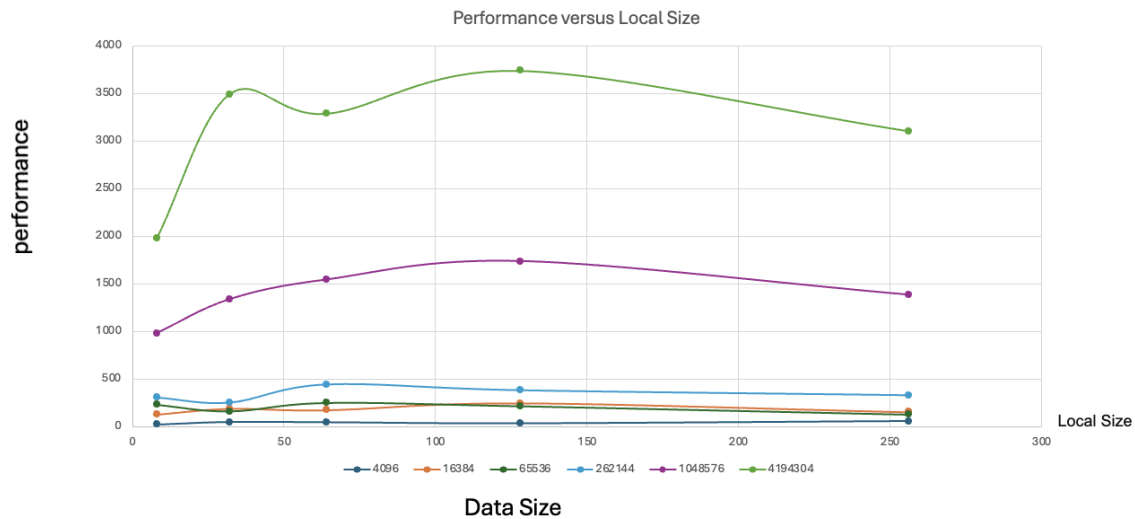
3. Show the table and graphs

performancne value	Data Size						
Local Size		4096	16384	65536	262144	1048576	4194304
8		28.36	134.14	235.39	311.35	983.15	1978.25
32		53.98	192.16	166.11	259.24	1338.9	3496.45
64		51.56	178.2	255.65	444.5	1546.54	3291.95
128		41.26	247.81	220.7	385.94	1741.38	3743.65
256		63.31	157.95	131.69	335.13	1387.39	3109.24

Graph 1: Performance versus DATASIZE, with a series of colored Constant-Local-Size curve.



Graph2: Performance versus Local Size (LOCALSIZE), with a series of colored Constant-DATASIZE curves



4. What patterns are you seeing in the performance curves? What difference does the size of the data make? What difference does the size of each work-group make?

Patterns in the performance curves:

From the graph 1, I found out that all five lines representing different local sizes (8, 32, 64, 128, and 256) show an upward trend, indicating that performance increases as data size increases. The initial part of each curve (from data size 0 to around 100000) shows a steep increase in performance. After this initial steep increase, the rate of performance gain slows down but continues to grow as data size increases. The line representing local size 256 consistently shows higher performance across all data sizes compared to the other local sizes.

From the graph 2, I found out that The green line (4194034 datasize) shows a peak in performance at a local size of around 100 before it starts to decline, and it also has the best performance.

Difference that the size of the data makes:

Larger data sizes generally lead to higher performance values for all work-group sizes and the impact of increasing data size appears more significant at smaller values; there's a sharp increase in performance when moving from small to moderate data sizes.

Difference that the size of each work-group makes:

Larger work-groups seem to offer better performance. For example, local size 256 outperforms smaller local sizes across almost all points on the graph. The gap between different work-group sizes appears more pronounced at larger data sizes.

5. Why do you think the patterns look this way?

There are many reasons why the patterns look this way. The size of the dataset (DATASIZE) and the work-group size (LOCALSIZE) significantly impact GPU performance. Larger datasets enable more parallelism, optimizing the GPU's computational capabilities and enhancing performance. However, this improvement has limits due to the overhead of managing large datasets and hardware constraints like memory bandwidth. Similarly, larger work-groups can improve performance by better utilizing the GPU's memory hierarchy and reducing scheduling overhead, but excessively large work-groups may cause resource contention and decrease performance. The optimal work-group size depends on the hardware's characteristics and the nature of the workload.